

6 User engagement analysis: analyze how users engage with content on social media to understand what types of content are most engaging

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from textblob import TextBlob
from wordcloud import WordCloud
```

```
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
True
```

```
df = pd.read_csv('location.csv')
df.head()
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 👏 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend	Positive	2023-01-15	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	

Next steps:

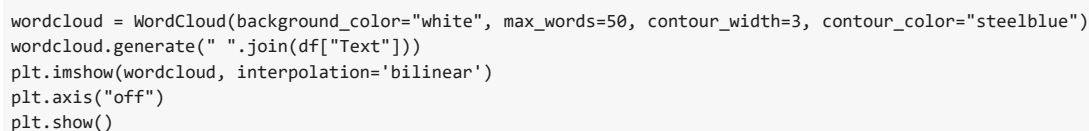
[Generate code with df](#)
[View recommended plots](#)
[New interactive sheet](#)

```
stop_words = set(stopwords.words('english'))
df["Text"] = df["Text"].apply(lambda x: " ".join(word for word in x.split() if word.lower() not in stop_words))
```

```
df["polarity"] = df["Text"].apply(lambda x: TextBlob(x).sentiment.polarity)
df.head()
```

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

Engagement Analysis of Tweets



```
sns.histplot(data=df, x=df["polarity"], bins=20)  
plt.xlabel("Polarity scores")  
plt.ylabel("Count")  
plt.title("Sentiment Analysis of Tweets")  
plt.show()
```

